

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
|   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| A |   |   |   |   |   |   |   |   |
| B |   |   |   |   |   |   |   |   |
| C |   |   |   |   |   |   |   |   |
| D |   |   |   |   |   |   |   |   |
| E |   |   |   |   |   |   |   |   |
| F |   |   |   |   |   |   |   |   |

Power input, soft-latch, supplies



File: power.kicad\_sch

3-phase power stage and current sensing



File: bridge.kicad\_sch

STM32G431 controller



File: mcu.kicad\_sch

Connectors and I/O conditioning



File: io.kicad\_sch

DESIGN SUMMARY

- Battery: 48 V nominal (13S Li-ion, 54.6 V full). Abs. max 60 V (TVS SMCJ60CA, 100 V FETs/drivers/bucks).
- Battery current limit 35 A (firmware, from FOC power estimate). Phase current limit 60 A (firmware), hard trip 80 A.
- Power stage: 6 x IPT015N10N5 (100 V, 1.5 mOhm, TOLL) driven by 3 x LM5109B half-bridge drivers @ 20 kHz, 4.7 Ohm gate resistors.
- Current sensing: in-line 0.5 mOhm 4-terminal (Kelvin) phase shunts + INA240A1 (G=20, mid-rail ref) -> +/-165 A full scale.
- Control: STM32G431CB @170 MHz, hall-sensor FOC with angle interpolation; hall/R/L auto-detect over UART.
- Supplies: LM5164 48->12 V (gate drive), LM5164 48->5 V (halls/throttle), AP2112 5->3.3 V (logic).
- Soft power: momentary button to GND wakes the bucks via Q1, MCU latches with PWR\_HOLD; long press = off.
- I/O: hall + motor NTC (JST-PH6), throttle (PH3), FWD/REV switch (PH2), brake (PH2), power button (PH2), UART 115200 8N1 3.3 V (PH4: 3V3, TX, RX, GND), SWD (1x5 header).

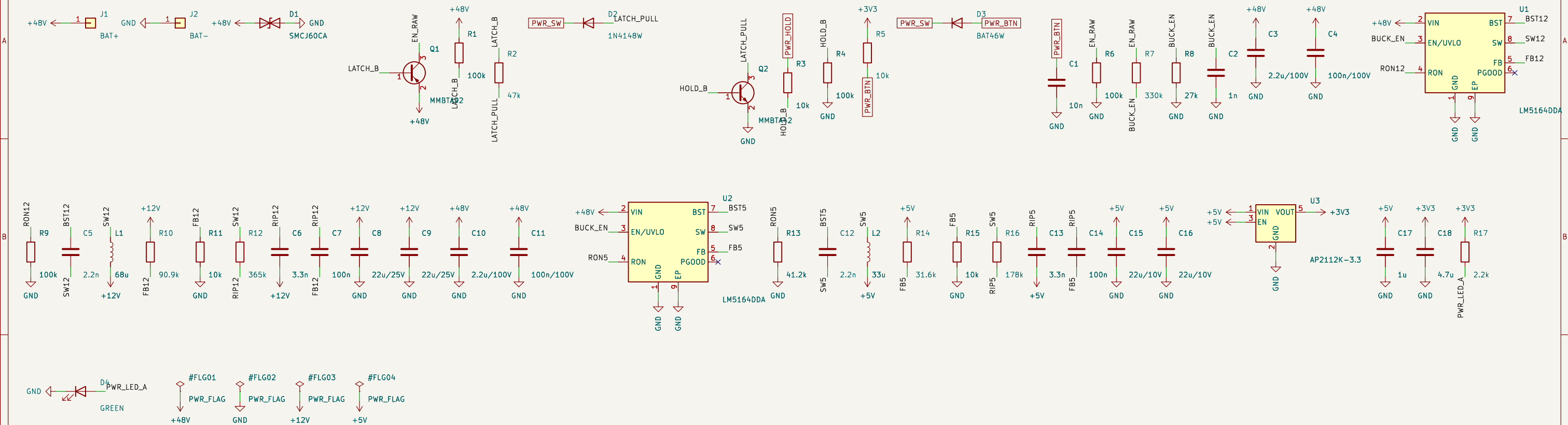
LAYOUT RULES: 2 oz copper min (4 layer preferred), bus/phase pours >= 10 mm wide, FET tabs on thermal-via arrays to an aluminium heat spreader, Kelvin-route INA240 inputs from the shunt pads, DC-link ceramics at each half bridge.

Top level  
STM32G431 + LM5109B x3 + IPT015N10N5 x6 + INA240A1 x3  
**48V 35A hub motor controller**  
Sheet: /  
File: bldc48.kicad\_sch

**Title: 48V 35A BLDC hub motor controller**

|                                               |                  |        |
|-----------------------------------------------|------------------|--------|
| Size: A3                                      | Date: 2026-10-06 | Rev: A |
| KiCad E.D.A.    kicad-cli 7.0.11+dfsg-1build4 | Id: 1/5          |        |

Power input, soft-latch, supplies

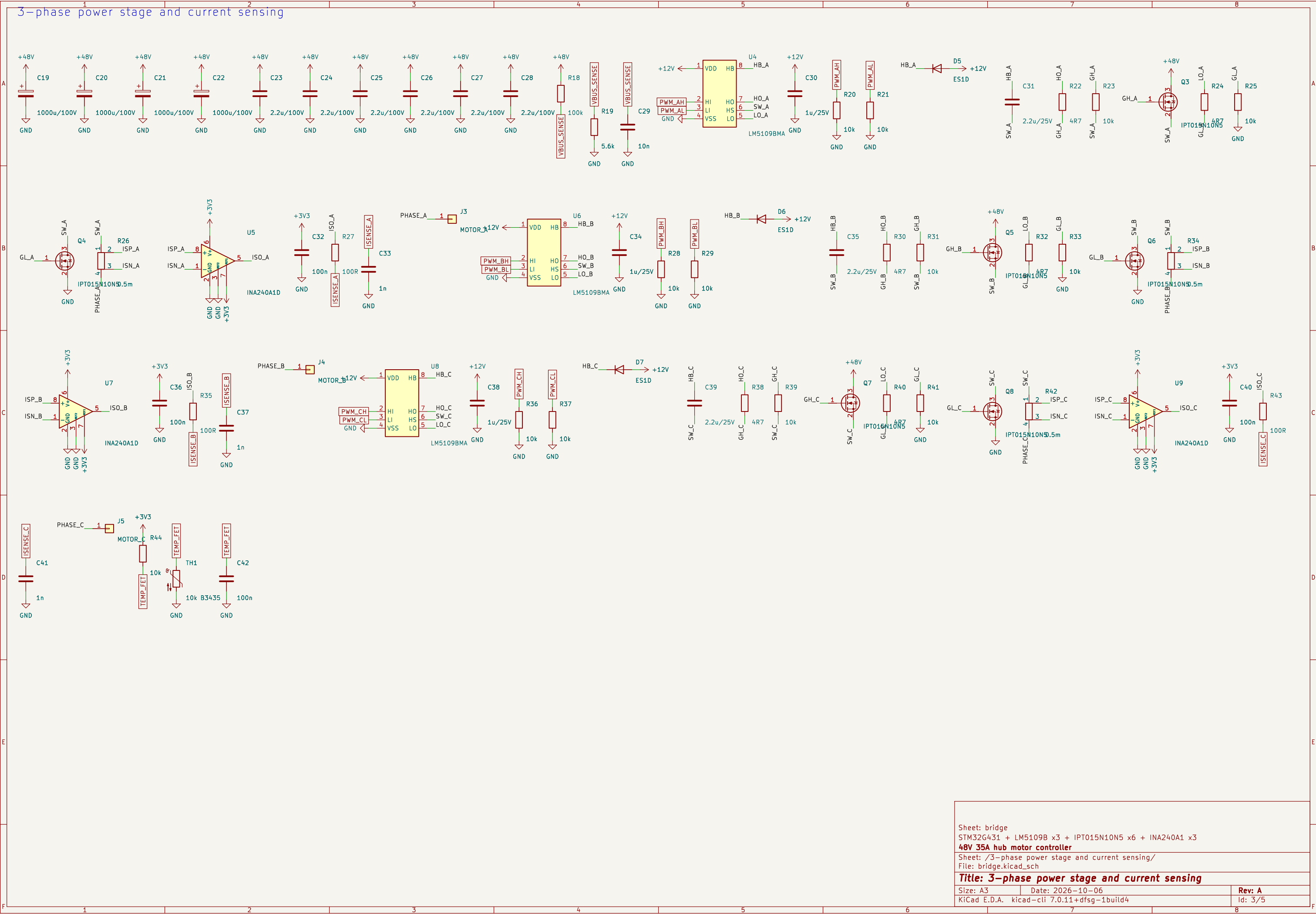


Sheet: power  
STM32G431 + LM5109B x3 + IPT015N10N5 x6 + INA240A1 x3  
**48V 35A hub motor controller**  
Sheet: /Power input, soft-latch, supplies/  
File: power.kicad\_sch

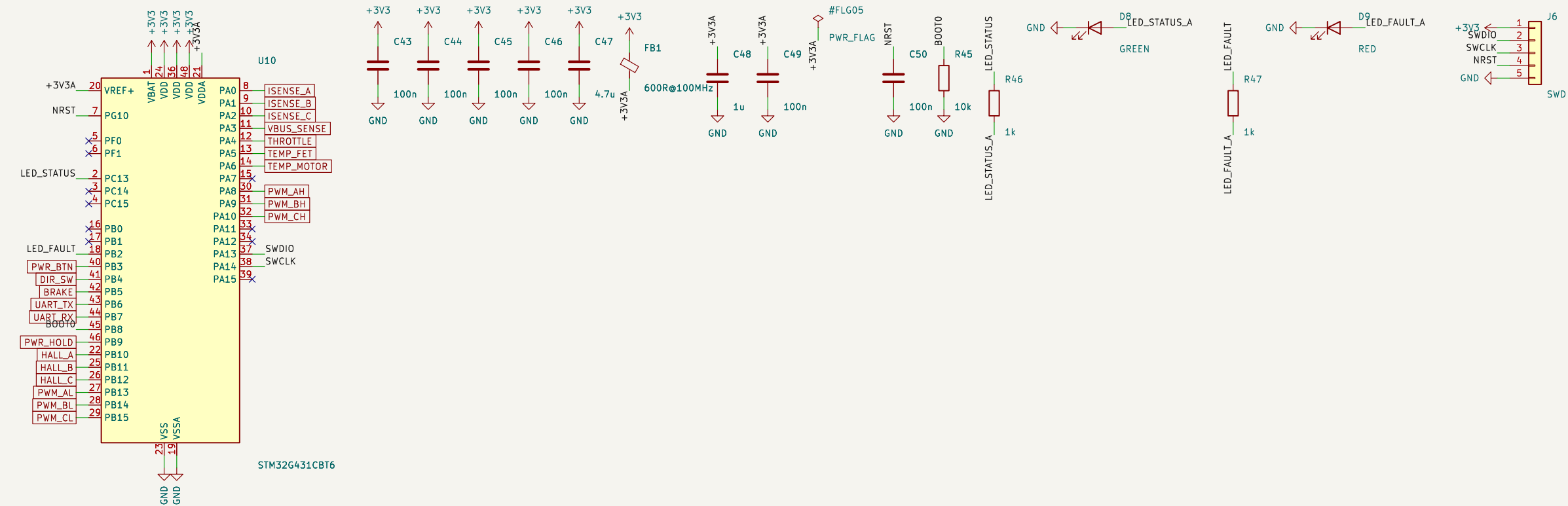
**Title: Power input, soft-latch, supplies**

|                                            |                  |         |
|--------------------------------------------|------------------|---------|
| Size: A3                                   | Date: 2026-10-06 | Rev: A  |
| KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 |                  | Id: 2/5 |

3-phase power stage and current sensing

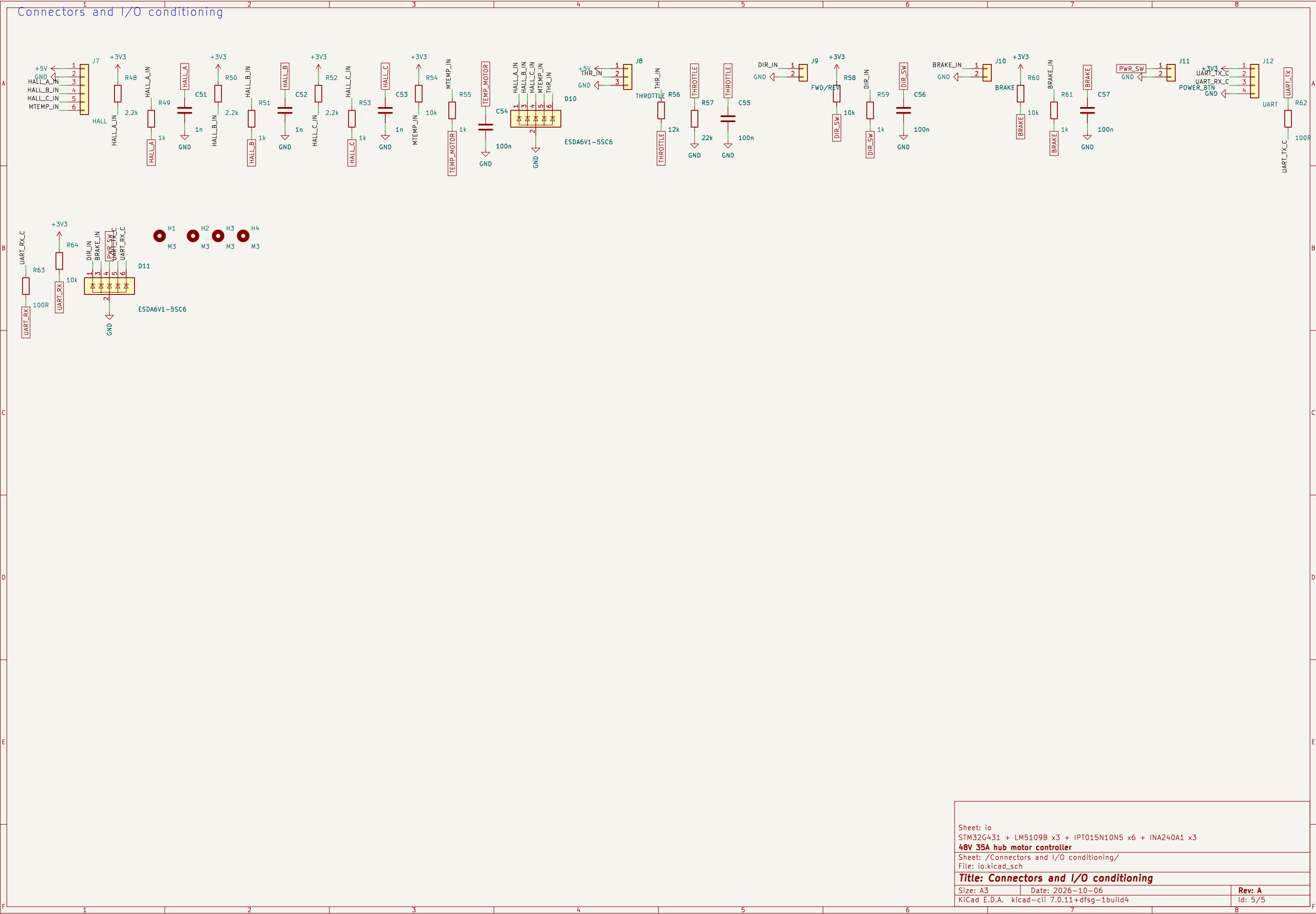


STM32G431<sup>1</sup> controller



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Sheet: mcu
STM32G431 + LM5109B x3 + IPT015N10N5 x6 + INA240A1 x3
48V 35A hub motor controller
Sheet: /STM32G431 controller/
File: mcu.kicad_sch
Title: STM32G431 controller
Size: A3 | Date: 2026-10-06
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
```

Connectors and I/O conditioning



Sheet: io  
STM32G431 + LM5109B x3 + IPT015N10N5 x6 + INA240A1 x3  
**48V 35A hub motor controller**  
Sheet: /Connectors and I/O conditioning/  
File: io.kicad\_sch

**Title: Connectors and I/O conditioning**

Size: A3 Date: 2026-10-06  
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: A  
Id: 5/5